Homework — 1.3.1 Compression, Encryption and Hashing — ANSWER SHEET

Separate answer sheet / mark scheme — keep for marking.

Mark scheme

Part A (14 marks)

1. [2] 1 mark: lossless compression. 1 mark for a valid reason, e.g. installers/program code and source files must be restored *exactly* / any single changed bit could corrupt the program / lossy would discard data and break the files. (No mark for choosing lossy.)

2. [2] Encoded output: `4R 3G 2R 4B` (or `R4 G3 R2 B4`). 1 mark for correct values in order (R, G, R, B); 1 mark for correct counts (4, 3, 2, 4).

3. [2] 1 mark: there are no runs of repeated values — every value differs from its neighbour, so each is stored as a count of 1. 1 mark: the encoded output (`1R 1G 1R 1G…`) is *larger* than the original because each pixel now needs both a value and a count.

4. [3] Up to 3 from: recurring patterns/strings/words are stored once in a dictionary/index **[1]**; each occurrence in the text is replaced by a short reference/index pointing to the dictionary entry **[1]**; because the reference is shorter than the repeated pattern the overall file is smaller **[1]**; the dictionary itself must be supplied/stored with the compressed data so it can be decompressed **[1]** (max 3).

5. [3] 1 mark: Bob's public key is shared openly / Alice obtains Bob's public key. 1 mark: Alice encrypts the file using Bob's public key. 1 mark: only Bob's matching private key (kept secret) can decrypt it, so even if the ciphertext is intercepted it cannot be read.

6. (a) [1] Hashing is one-way / not reversible — the original password cannot be calculated from the stored hash. (Accept: even with the hash an attacker cannot recover the password.)

(b) [1] At login the entered password is hashed using the same hash function and the result is compared with the stored hash; if they match the password is correct.

Part B (6 marks)

7. [2] 1 mark + 1 mark for a clear difference, e.g. a compiler translates the *whole* program at once into object/machine code before execution, producing an executable **[1]**; an interpreter translates and executes one line/statement at a time, with no standalone executable produced **[1]**. (Accept: compiler reports all errors at end vs interpreter stops at first error; compiled code runs faster.)

8. [2] 1 mark: the **MAR** holds the address of the memory location to be read (the address of the next instruction, copied from the PC). 1 mark: the **MDR** holds the data/instruction that has been fetched from that memory location.

9. [2] 1 mark: the scheduler decides which process/task gets CPU time / allocates processor time to processes. 1 mark: to manage multiple processes fairly/efficiently (e.g. maximise CPU use, avoid starvation) — accept naming a scheduling aim or algorithm (round robin, etc.).

Total: 14 + 6 = 20 marks